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Studierichting: Informatica
Oefeningenreeks 2B nr. 5.2

$$\begin{aligned}\text{Los op: } \lim_{x \rightarrow 0} \left(\frac{1}{\sqrt{2-x} - \sqrt{x+2}} \right) \\ \lim_{x \rightarrow 0^+} \left(\frac{1}{\sqrt{2-x} - \sqrt{x+2}} \right) &= \frac{1}{\sqrt{2-0^+} - \sqrt{0^++2}} \\ &= \frac{1}{\sqrt{2^-} - \sqrt{2^+}} = \frac{1}{0^-} = -\infty\end{aligned}$$

en:

$$\begin{aligned}\lim_{x \rightarrow 0^-} \left(\frac{1}{\sqrt{2-x} - \sqrt{x+2}} \right) &= \frac{1}{\sqrt{2-0^-} - \sqrt{0^-+2}} \\ &= \frac{1}{\sqrt{2^+} - \sqrt{2^-}} = \frac{1}{0^+} = +\infty\end{aligned}$$

References